

Paper 4B: Further Statistics 2

Topics	What students need to learn:		
	Content		Guidance
1 Linear Regression	1.1	Least squares linear regression. The concept of residuals and minimising the sum of squares of residuals.	Students should have an understanding of the process involved in linear regression. They should be able to calculate the regression coefficients for the equation y on x using standard formulae.
	1.2	Residuals. The residual sum of squares (RSS)	An intuitive use of residuals to check the reasonableness of linear fit and to find possible outliers. Use in refinement of mathematical models. The formula $RSS = S_{yy} - \frac{(S_{xy})^2}{S_{xx}}$ Derivations are not required